

CS208 Cloud Computing and Cloud Networking

Assignment 1 (Due 11:59 pm, 23 Jan 2025)

Virtual Machine

The goal of this assignment is to give an introduction to Virtual machines.

Problem Statement

In this assignment, you will create two virtual machines and write a client-server socket program. You will run a server program on one VM and a client program on another VM. Once the client establishes a connection with the server, the server will send a file from the local filesystem, and the client will save it on the local system.

Server Application

1. Create a file “mydata.txt” of any size with random content. Note down its checksum.
2. When you run the server application, it will read a file “mydata.txt” from the local filesystem.

Client Application

1. The client application connects to the server running on a different VM, receives the file that the server sends, and saves it in its local filesystem “mydata_client_copy.txt”.
2. Open another terminal and verify that the file “mydata_client_copy.txt” is received properly by checking the checksum.

General instructions

- The client and server application itself can be built using any language you are comfortable with. You can use Virtual Box to create two VMs locally on your laptop
- or use Cloudlab experiment with the small-lan profile and use ubuntu-20.04 image and check VM box.

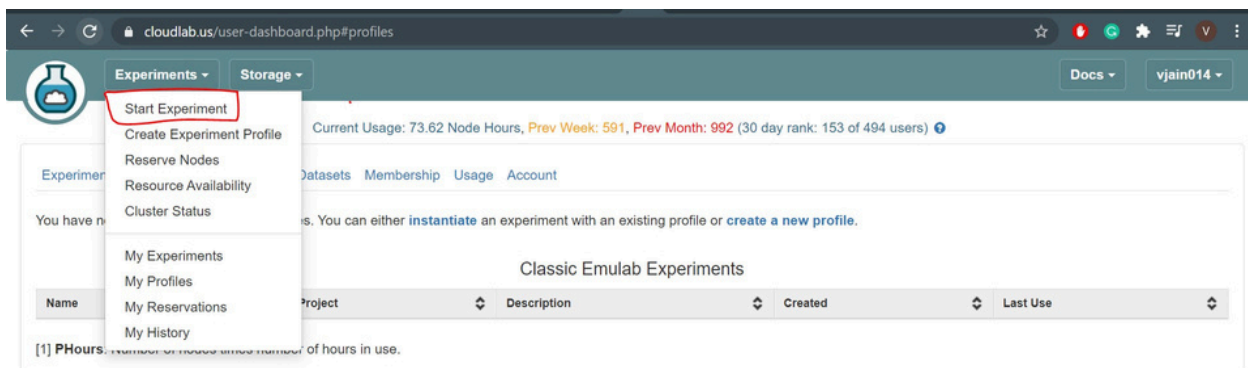
Grading

Grade Distribution

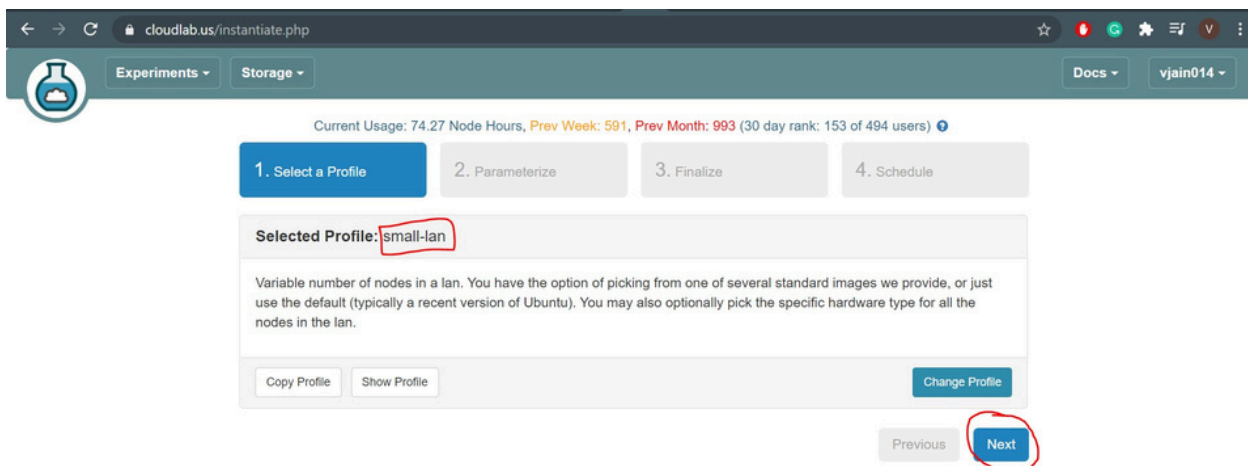
- Server and client programs - 30
- Screenshot of server application running on VM 1- 25 points
- Screenshot of client application running on VM 2- 25 points
- Checksum verification - 10 points
- Steps and screenshot to reproduce the experiment- 10 points

Creating VMs on Cloudlab

1. Make sure when you create the cloudlab account, you provide your public ssh key.
2. Login to the cloudlab portal and start an experiment



3. Use small-lan profile



4. Enter the number of VMs, OS version and Check “Use XEN VMs”

The screenshot shows the 'Parameterize' step of the CloudLab experiment instantiation process. The browser address bar shows 'cloudlab.us/instantiate.php'. The page has a progress bar with four steps: 1. Select a Profile, 2. Parameterize (active), 3. Finalize, and 4. Schedule. Below the progress bar, a message states: 'This profile is parameterized; please make your selections below, and then click Next.' There are tabs for 'Resource Availability', 'Defaults', 'Last', and 'History'. A '+ Show All Parameter Help' link is present. The parameters are: 'Number of Nodes' set to 2, 'Select OS image' set to 'UBUNTU 20.04', 'Optional physical node type' is empty, and 'Use XEN VMs' is checked. An 'Advanced' link is at the bottom left. 'Previous' and 'Next' buttons are at the bottom right.

Current Usage: 74.27 Node Hours, Prev Week: 591, Prev Month: 993 (30 day rank: 153 of 494 users)

1. Select a Profile 2. Parameterize 3. Finalize 4. Schedule

This profile is parameterized; please make your selections below, and then click **Next**.

[+ Show All Parameter Help](#)

Number of Nodes 2

Select OS image UBUNTU 20.04

Optional physical node type

Use XEN VMs ☒

[Advanced](#)

[Previous](#) [Next](#)

5. Press Next and select any cluster

6. After starting the experiment, you will get login details about VMs

The screenshot shows the 'status.php' page for a specific experiment. The browser address bar shows 'cloudlab.us/status.php?uid=75015869-50bf-11eb-b1eb-e4434b2381fc'. A green banner at the top says 'Your experiment is ready!'. Below this, experiment details are listed: Name: CS208assignment, State: ready, Profile: small-lan, Creator: vjain014, Project: KKKProjects, Created: Jan 7, 2021 12:08 AM, Started: Jan 7, 2021 12:08 AM, Expires: Jan 7, 2021 4:08 PM (in 15 hours). There are buttons for 'Logs', 'Performance History', 'Create Disk Image', 'Copy', 'Extend', and 'Terminate'. A 'Profile Instructions' link is also present. Below this is a table with tabs for 'Topology View', 'List View', 'Manifest', 'Graphs', and 'Bindings'. The 'List View' tab is active, showing a table of VMs.

cloudlab.us/status.php?uid=75015869-50bf-11eb-b1eb-e4434b2381fc

▼ Your experiment is ready!

Name: CS208assignment
State: ready
Profile: small-lan
Creator: vjain014
Project: KKKProjects
Created: Jan 7, 2021 12:08 AM
Started: Jan 7, 2021 12:08 AM
Expires: Jan 7, 2021 4:08 PM (in 15 hours)

[Logs](#) [Performance History](#) [Create Disk Image](#) [Copy](#) [Extend](#) [Terminate](#)

[Profile Instructions](#)

Topology View List View Manifest Graphs Bindings

ID	Node	Type	Status	Startup	Image	SSH command (if you provided your own key)	☐	Actions
vm0	c220g1-030802vm-1	pcvm	ready	n/a	emulab-ops/UBUNTU18-64-STD	ssh -p 26410 vjain014@c220g1-030802.wisc.cloudlab.us	☐	Settings
vm1	c220g1-030802vm-2	pcvm	ready	n/a	emulab-ops/UBUNTU18-64-STD	ssh -p 26411 vjain014@c220g1-030802.wisc.cloudlab.us	☐	Settings
c220g1-030802	c220g1-030802	c220g1	n/a	n/a	n/a	ssh -p 22 vjain014@c220g1-030802.wisc.cloudlab.us	☐	Settings